

Weekly training log, week 9

Module EEEM068, Applied Machine Learning

Project Industrial waste classification on WaRP C

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Focus Swin Tiny with self supervised pretraining using DINO and MIM

1. Experimental settings

Item	Value
Dataset	WaRP C
Training images	8767 images used for SSL dataset
Classes	28
Backbone	Swin Tiny patch 4 window 7 input 224
Metric	Top 1 accuracy and hard class recovery
Baseline for comparison	Swin optimised baseline at 81.56 percent

2. Trial summary

Trial	Method	Training setup	Best test accuracy
1	DINO SSL	30 SSL epochs. Then frozen and full fine tuning.	70.08 percent
2	MIM SSL	30 SSL epochs. Then full fine tuning.	79.69 percent
Reference	Swin baseline	Previously optimised supervised model.	81.56 percent

3. DINO SSL trial

The DINO trial used self distillation with a student and teacher Swin backbone. The notebook reduced the view count to two global and two local views for memory control. The pretrained backbone was then transferred to the classifier.

Item	Value
SSL epochs	30
SSL output dimension	4096
Student temperature	0.1
Teacher temperature	0.04
Teacher update	EMA
Fine tuning phase 1	Backbone frozen for 15 epochs
Fine tuning phase 2	Backbone unfrozen until epoch 60
Loss for fine tuning	Cross entropy with hard class weights

DINO pretraining log

Epoch	Loss	Learning rate	EMA momentum
1	6.7636	4.99e 05	0.9960
2	4.8342	9.99e 05	0.9960
3	2.3143	1.50e 04	0.9961
4	1.8937	2.00e 04	0.9962
5	1.7448	2.50e 04	0.9963
6	1.7011	3.00e 04	0.9964
7	1.7183	3.50e 04	0.9965
8	1.6918	4.00e 04	0.9967
9	1.6718	4.50e 04	0.9968
10	1.6549	5.00e 04	0.9970
11	1.6259	4.97e 04	0.9972
12	1.5883	4.88e 04	0.9974
13	1.5205	4.73e 04	0.9976
14	1.4949	4.52e 04	0.9978
15	1.4342	4.27e 04	0.9980
16	1.3985	3.97e 04	0.9982
17	1.3431	3.64e 04	0.9984
18	1.3130	3.27e 04	0.9986
19	1.2468	2.89e 04	0.9988
20	1.1965	2.50e 04	0.9990
21	1.1772	2.11e 04	0.9992
22	1.1283	1.73e 04	0.9993
23	1.0778	1.37e 04	0.9995
24	1.0398	1.03e 04	0.9996
25	1.0357	7.33e 05	0.9997
26	0.9916	4.78e 05	0.9998
27	0.9686	2.73e 05	0.9999
28	0.9617	1.23e 05	1.0000
29	0.9570	3.09e 06	1.0000
30	0.9395	1.03e 11	1.0000

DINO classifier fine tuning log

Epoch	Loss	Train accuracy	Test accuracy
1	2.9819	23.56	4.26
2	2.2523	33.04	10.64
3	2.0958	35.62	20.76
4	2.0411	38.59	28.56
5	1.9899	39.86	30.56
6	1.9486	40.22	32.62
7	1.9822	40.89	34.56
8	1.9343	42.06	35.40
9	1.9477	41.29	35.85
10	1.8903	41.44	35.98
11	1.9387	42.45	35.91
12	1.9148	41.31	36.04
13	1.9206	42.31	35.91
14	1.9344	41.71	35.91
15	1.9225	42.24	36.04
16	1.7973	52.42	37.14
17	1.6473	56.89	40.94
18	1.5524	63.37	44.36
19	1.4618	64.78	47.00
20	1.4125	66.91	50.29
21	1.3556	68.32	52.48
22	1.2805	70.43	55.19
23	1.2716	71.27	56.67
24	1.2078	73.58	58.61
25	1.2027	75.07	59.32
26	1.1790	74.98	59.90
27	1.1606	76.88	60.48
28	1.1272	76.66	61.70
29	1.1211	78.09	62.54
30	1.0986	77.39	63.31
31	1.0854	78.27	63.70
32	1.0731	80.36	64.54
33	1.0440	79.67	65.31
34	1.0194	80.76	65.51
35	1.0349	81.56	66.34
36	1.0154	81.74	66.60
37	1.0097	81.95	66.92
38	1.0118	82.31	67.38
39	0.9966	83.10	67.63
40	0.9802	83.86	68.09
41	0.9755	83.96	68.02
42	0.9701	83.02	68.28
43	0.9667	84.83	68.28
44	0.9525	84.18	68.34
45	0.9473	84.79	68.41
46	0.9338	85.04	68.79
47	0.9481	85.55	69.12
48	0.9375	86.05	69.25
49	0.9184	85.90	69.37
50	0.9267	84.98	69.76
51	0.9254	85.97	69.57
52	0.9365	86.26	69.44
53	0.9222	86.45	69.76
54	0.9249	86.77	69.89
55	0.9232	86.89	70.08
56	0.9339	85.98	70.02

Epoch	Loss	Train accuracy	Test accuracy
57	0.9190	87.16	69.89
58	0.9379	86.80	69.95
59	0.9236	86.24	70.02
60	0.9284	86.45	70.08

DINO result

Measure	Value
Best test accuracy	70.08 percent
Swin baseline	81.56 percent
Difference from baseline	11.48 percentage points lower

4. MIM SSL trial

The MIM trial used masked image modelling with a Swin encoder and a lightweight reconstruction decoder. The model reconstructed only the masked patches. The pretrained backbone was transferred to the classifier and trained with full fine tuning from the first epoch.

Item	Value
SSL epochs	30
Mask ratio	75 percent
Grid size	7 by 7 patches
Masked patches	36 of 49 patches
MIM loss	Mean squared error on masked patches
Fine tuning	Full fine tuning for 40 epochs

MIM pretraining log

Epoch	MIM loss
1	0.7363
2	0.6457
3	0.5961
4	0.5232
5	0.4503
6	0.4136
7	0.3800
8	0.3683
9	0.3519
10	0.3338
11	0.3195
12	0.3164
13	0.3069
14	0.2994
15	0.2971
16	0.2909
17	0.2835
18	0.2830
19	0.2786
20	0.2762
21	0.2737
22	0.2716
23	0.2690
24	0.2725
25	0.2682
26	0.2682
27	0.2667
28	0.2631
29	0.2673
30	0.2634

MIM classifier fine tuning log

Epoch	Loss	Train accuracy	Test accuracy
1	2.9910	7.60	2.84
2	2.6468	15.77	2.84
3	2.3946	25.70	4.26
4	2.1768	37.65	5.35
5	1.9728	46.17	7.61
6	1.7382	54.97	13.22
7	1.5615	62.19	21.21
8	1.3641	73.18	28.37
9	1.2297	82.90	35.33
10	1.1215	86.64	45.13
11	1.0024	88.78	56.03
12	0.9386	90.43	64.28
13	0.9083	92.22	69.95
14	0.8780	92.66	72.40
15	0.8686	94.37	75.11
16	0.8301	94.61	76.34
17	0.8062	94.32	77.05
18	0.7953	95.81	77.43
19	0.7904	95.49	77.63
20	0.7838	96.18	77.82
21	0.7740	96.42	78.21
22	0.7447	96.59	78.53
23	0.7446	96.79	78.59
24	0.7382	96.93	78.66
25	0.7415	97.45	78.72
26	0.7550	97.04	78.92
27	0.7356	97.23	79.05
28	0.7355	97.49	79.30
29	0.7383	97.62	79.56
30	0.7312	97.76	79.63
31	0.7116	97.53	79.43
32	0.7276	97.81	79.50
33	0.7281	97.64	79.37
34	0.7198	97.57	79.43
35	0.7200	97.79	79.37
36	0.7117	97.83	79.37
37	0.7164	97.86	79.43
38	0.7270	97.83	79.43
39	0.7300	97.66	79.63
40	0.7096	97.94	79.69

MIM result

Measure	Value
Best test accuracy	79.69 percent
Swin baseline	81.56 percent
Difference from baseline	1.87 percentage points lower

5. Final comparison

Model	Top 1 accuracy
Baseline Swin	81.56 percent
DINO	70.08 percent
MIM	79.69 percent

Hard class recovery

Class	Baseline	DINO	MIM
bottle oil full	37.5 percent	37.5 percent	62.5 percent
bottle multicolor	46.4 percent	57.1 percent	53.6 percent
bottle multicolorv full	61.9 percent	61.9 percent	66.7 percent
bottle blue5l full	58.3 percent	83.3 percent	58.3 percent
detergent transparent	65.9 percent	68.3 percent	70.7 percent

Observation

DINO improved some hard classes but reduced the overall top 1 accuracy. MIM gave the strongest self supervised result and came close to the supervised Swin baseline. The MIM result is useful for the report because it shows hard class recovery without beating the best fully supervised baseline.